

2nd Law of Thermodynamics.

①

Equivalent statements:

1) Kelvin-Planck statement

It is impossible to construct an engine which operating in a cycle, will produce no other effect than extraction of heat from a reservoir and the performance of an equivalent amount of work.

2) Clausius statement:

It is impossible to construct a device which operating in a cycle, will produce no other effect than transfer of heat from a colder to hotter body.

These two statements are equivalent.

In this regard, another state function (which can be considered as a co-ordinate like P, V, T) "entropy" has been introduced.

Entropy:

The state function entropy (S) can be mathematically defined as.

$$dS = \frac{\delta Q}{T}$$

where δQ is the exchange of heat and T is the temperature.

Point to be noted that although δQ is path dependent funcⁿ (hence not a state funcⁿ), dS is a state funcⁿ.

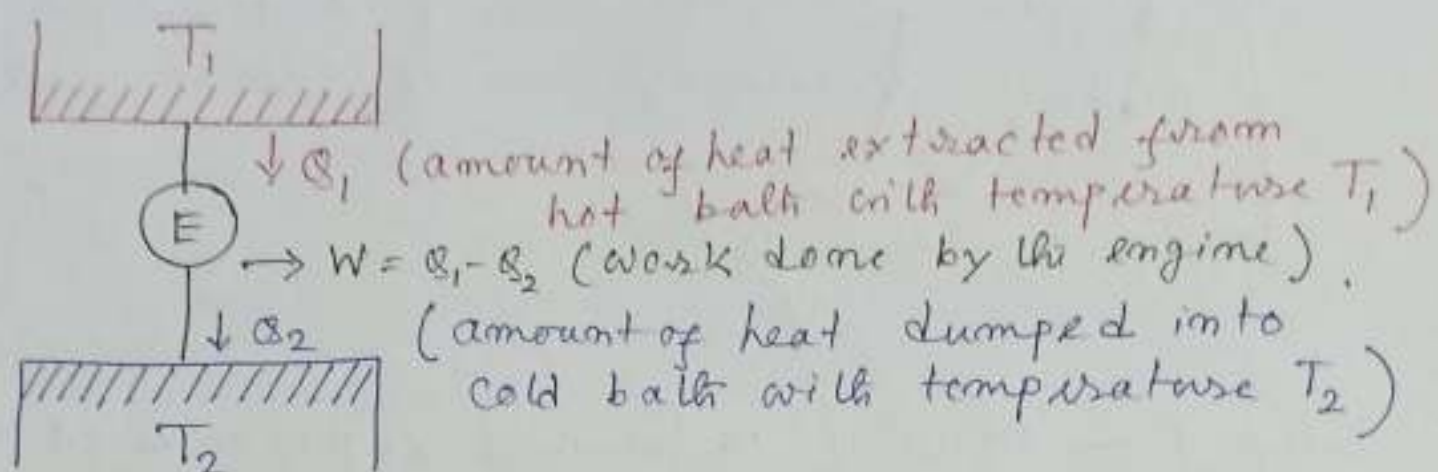
$$S = \int_A^B dS = \int_A^B \frac{\delta Q}{T} = S_B - S_A$$

depending only on the entropies at B and A .

2nd law of thermodynamic can therefore be expressed as - "The entropy of the universe (i.e. system + environment) always increases.

$$\Delta S \geq 0. \quad [\text{Equality holds for reversible processes}]$$

There is a lot of physics involved here which has been omitted for the purpose of this Math. elective. Interested people should see "Clausius inequality".

Carnot cycle :

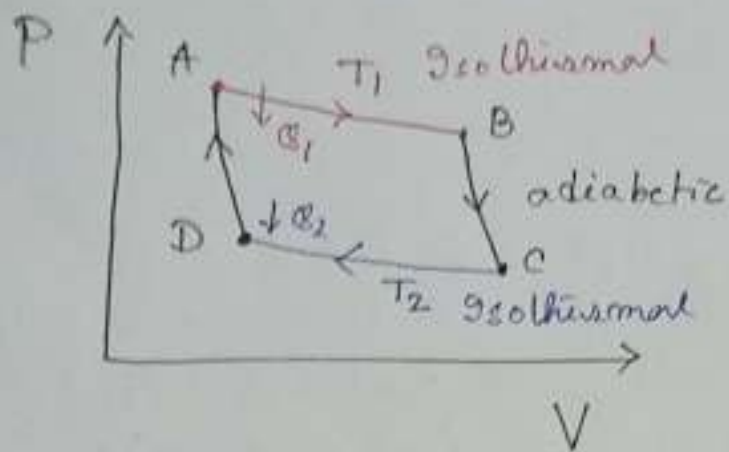
Above schematic is for a basic heat engine which performs in between two heat baths with temperature T_1 and T_2 ($T_1 > T_2$)

The efficiency of the engine

$$\eta = \frac{W}{Q_1} \left(\frac{\text{Work done}}{\text{heat extracted}} \right)$$

$$\eta = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

We now discuss a reversible heat engine in the form of a cyclic process called Carnot cycle. It consists of four processes two of which are adiabatic and two are isothermal.



P-V diagram
of Carnot cycle.

Step-1 - It is a isothermal expansion ($A \rightarrow B$) of the working substance, where Q_1 amount of heat is extracted from hot bath.

Step-2 - It is an adiabatic expansion ($B \rightarrow C$)
Since $\Delta Q = 0$, $\Delta S = 0$ for this process

Step-3 - It is a isothermal compression ($C \rightarrow D$) where Q_2 amount of heat is dumped into the cold reservoir.

Step-4 - It is a adiabatic compression ($D \rightarrow A$) resetting the medium to its initial state, hence the completion of the cycle.

(5)

$$Q_1 = m R T_1 \ln\left(\frac{V_2}{V_1}\right) \quad (A \rightarrow B)$$

$$W_{B \rightarrow C} = \frac{m R}{\gamma - 1} (T_1 - T_2) \quad (B \rightarrow C)$$

$$Q_2 = m R T_2 \ln\left(\frac{V_3}{V_4}\right) \quad (C \rightarrow D)$$

$$W_{D \rightarrow A} = \frac{m R}{\gamma - 1} (T_2 - T_1).$$

$$W = m R T_1 \ln\left(\frac{V_2}{V_1}\right) - m R T_2 \ln\left(\frac{V_3}{V_4}\right)$$

$$[\text{Since } W_{B \rightarrow C} = -W_{D \rightarrow A}]$$

$$\frac{V_2}{V_3} = \frac{V_1}{V_4} \quad [\text{check using adiabatic eqn}].$$

So, efficiency

$$\eta = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

No ~~of~~ engine is more efficient than reversible engines. Hence no engine is more efficient than Carnot engine.
(Another 2nd law statement)

Three T-dS equations:

From 1st law we know.

$$dQ = dU + PdV \quad [\text{For hydrodynamic systems}]$$

$$= C_V dT + PdV. \quad \text{---}$$

Using 2nd law.

$$TdS = C_V dT + PdV. \quad \text{--- (1)}$$

For Ideal gas $C_p - C_V = mR$

$$PdV + VdP = mR dT.$$

$$\text{hence} \rightarrow TdS = C_p dT - VdP \quad \text{--- (2).}$$

further (if we eliminate $\frac{dT}{dT}$ from, then ^{right side of} the eqnⁿ).

$$TdS = \frac{1}{mR} (C_p PdV + C_V VdP).$$

Rearranging $\rightarrow$

$$dS = C_p \frac{dV}{V} + C_V \frac{dP}{P} \quad \text{--- (3)}$$